Hamza Shaikh

Site Reliability · DevOps · Cloud Infrastructure

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Summary

Site Reliability and DevOps engineer, currently completing an M.Sc. in Computer Science at RPTU Kaiserslautern with major tracks in Distributed Systems and Software Engineering. Sole infrastructure engineer at a D2C startup: migrated production from EC2 to ECS/Fargate, owned the VPC, security-group and IAM layer, built the CI/CD pipeline and ran incident response. Builds reliability tooling in Go and Python, including a chaos engineering platform and a validator anomaly detector that placed 3rd at the Superteam Germany ideathon in Frankfurt. Interested in observability, failure testing, and the question of whether monitoring actually fires on the things it was built to catch.

Location & Work Eligibility

Currently based: Kaiserslautern, Germany
Germany: German student residence permit. Work authorisation for 140 full or 280 half days per year. **No visa sponsorship required.**
Saudi Arabia: Valid dependent iqama under Saudi Aramco sponsorship; eligible for an HRSD work permit with no block visa quota required. Resident in Dammam, Eastern Province since 2007. Valid Saudi driving licence (to 2030).
Elsewhere: Visa sponsorship required. Currently on a German student residence permit, Indian passport.
India: Indian national, unrestricted right to work.
Open to: Werkstudent, internship and junior F/T roles in Germany; full-time roles in Saudi Arabia and India and Remote.
Languages: English (professional), Hindi/Urdu, Arabic (conversational), German (A2, progressing)

Technical Skills

Cloud & Networking: AWS (EC2, ECS, Fargate, VPC, Subnets, Security Groups, IAM, S3, CloudWatch, Load Balancing), Cloudflare, DigitalOcean
Infrastructure: Docker, Docker Compose, Linux, Terraform, NGINX, container hardening, Kubernetes (foundational)
SRE & Observability: Site Reliability Engineering, Prometheus, PromQL, Grafana, Alertmanager, RED metrics, SLOs, chaos engineering, monitoring and alerting, incident response
CI/CD & Security: Git, GitHub Actions, CI/CD pipelines, Bash, Trivy, Bandit, pip-audit, govulncheck, pytest
Programming & Data: Python, Go, C++, FastAPI, Redis, BullMQ, MySQL

Experience

Patil Kaki, D2C Food Brand (Shark Tank India) — **Jun. 2023 – Aug. 2023**
DevOps Engineering Intern (Sole Infrastructure Owner) — *Mumbai, India (Remote)*

- Migrated production from EC2 to ECS Fargate as the only infrastructure engineer in a 10-person team, supporting a storefront at ~2,500 orders/month; dynamic task provisioning cut AWS spend 27% ($230 to $168/month).
- Built the AWS network and access layer: VPC subnets, routing and security groups; S3 storage; least-privilege IAM roles scoped separately for founders, developers and logistics staff. Fronted the platform with Cloudflare.
- Root-caused a three-day chatbot outage: a whitespace character intermittently injected into the DigitalOcean-generated integration URL was silently breaking the widget. Traced it through the hosting integration and restored the service.
- Built GitHub Actions pipelines gating main-branch deployments with rollback, supporting weekly releases.
- Decoupled order processing from the request path using BullMQ over Redis, moving fulfilment work off the synchronous checkout flow.

CMP Infotech (Microsoft Partner Program) — **Dec. 2021 – Jan. 2022**
Java Development Intern — *Mumbai, India (Remote)*

Projects

AutoSRE: Chaos Engineering Platform | *Python, Go, FastAPI, Prometheus, Docker, GitHub Actions* **Feb. 2026 – Jul. 2026**

- Built a chaos engineering framework injecting latency, errors, CPU burn and memory pressure into live services through a REST control plane. Experiments are bounded, time-limited and self-reverting; the control plane, health probes and metrics endpoints are exempt from injection so the abort path can never be made unreachable.
- Made experiments reproducible rather than random: errors are spread deterministically across requests, and the memory injector touches one byte per page to force real commit instead of a lazy allocation.
- Implemented a three-state circuit breaker (closed, open, half-open) with an injectable clock for deterministic tests, a pluggable store interface and Prometheus state metrics, plus token-bucket rate limiting middleware returning 429 with Retry-After.
- Instrumented RED metrics through ASGI middleware with bounded label cardinality; defined a p95 500ms SLO with a 750ms alert threshold as Prometheus rules, unit-tested to assert every alert carries severity and a runbook link.
- Hardened the delivery path: 9-job CI running Bandit, Trivy, pip-audit and govulncheck with third-party actions pinned to commit SHAs; containers run read-only and non-root with all capabilities dropped and a distroless image for the Go daemon.
- 93.9% line coverage on the shared platform library. Documented with architecture decision records and game-day postmortems, each carrying a command to reproduce the incident.

SentinelSOL: Distributed Validator Node Monitoring | *Go, Prometheus, PromQL, Alertmanager, Grafana* **Sep. 2025 – May 2026**

- Built a Go telemetry daemon polling a Solana validator's JSON-RPC out of process into Prometheus; backoff retries only transient failures, and both endpoints scrape concurrently under a WaitGroup so the metrics share a timestamp.

- Implemented 3-sigma Z-score anomaly detection on vote velocity as Prometheus recording and alerting rules against a rolling one-hour baseline learned from the node's own history, routed through Alertmanager to Telegram.
- Fixed a detection blind spot: the original rate-ratio rule went undefined as the validator stalled, silencing the alert at exactly the moment it existed to catch.
- Packaged the full telemetry stack with Docker Compose for single-command bare-metal deployment, with Grafana dashboards provisioned as code.
- **Winner**, Superteam Germany / neosfer Solana Ideathon, Frankfurt (2026). Also submitted to the Colosseum Frontier hackathon.

GridCast: Heatwave Forecasting | *Python, TensorFlow, Keras, LSTM, Flask, React.js, Optuna* **Aug. 2023 – May 2024**

- Built 28 region- and season-specific LSTM models (7 regions x 4 seasons) trained on 70 years of IMD gridded temperature data (1951–2021) for India-wide heatwave forecasting.
- Designed a 14-day sliding-window input pipeline predicting 7 days ahead, with Optuna (Bayesian TPE) hyperparameter search across all 28 model subsets.
- Deployed end to end as a Flask REST API serving model inference behind a React.js frontend with D3.js geospatial visualisation. B.E. capstone project, submitted to IEEE ICSPCRE 2024.

Education

RPTU Kaiserslautern-Landau **Oct. 2025 – Expected 2027**
M.Sc. Computer Science (Specialisation 1: Distributed Systems, 2: Software Engineering) *Germany*

- Coursework: Distributed Systems, Software Engineering, Machine Learning.

Career Break & M.Sc. Admissions **Jun. 2024 – Sep. 2025**
Post-graduation period with family in Dammam; graduate applications and German student visa process *Intake deferred to Oct. 2025*

Xavier Institute of Engineering **Aug. 2020 – Jun. 2024**
*B.E. Information Technology | CGPA **8.69 / 10** (German equivalent: **1.6**)* *Mumbai, India*

International Indian School Dammam **Aug. 2010 – 2018**
CBSE Schooling *Dammam, Saudi Arabia*

Certifications & Community

Certifications: AWS Certified Cloud Practitioner (2023); Postman API Fundamentals Student Expert (2023)
Community: Hack The League, Mumbai and Hack This Fall, Pune (2022); Solana Hacker House (BLR / MUM 2023, BLR 2024); Berlin Blockchain Week, Solana Summit Germany and ETH Day, Berlin (2026)